

Experiment 20

Understand the K-means clustering algorithm for a given data set.

Program:

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
%matplotlib inline

df=pd.read_csv('Mall_Customers.csv')
df.info()
df.head()

sns.pairplot(df)

features=df.iloc[:, [3,4]].values

from sklearn.cluster import KMeans
model=KMeans(n_clusters=5)
model.fit(features)

Final=df.iloc[:, [3,4]].copy()
Final['label']=model.predict(features)
Final.head()

sns.set_style("whitegrid")
sns.FacetGrid(Final,hue="label",height=8) \
.map(plt.scatter,"Annual Income (k$)",
     "Spending Score (1-100)") \
.add_legend()
plt.show()

# Elbow method
features_el=df.iloc[:, [2,3,4]].values
wcss=[]
for i in range(1,10):
    model=KMeans(n_clusters=i)
    model.fit(features_el)
    wcss.append(model.inertia_)

plt.plot(range(1,10),wcss)
plt.show()
```

Output

Output:
Dataset shape: 200 rows × 5 columns.

KMeans(n_clusters=5)

First five clustered records:

	Annual Income (k\$)	Spending Score (1-100)	label
0	15	39	4
1	15	81	2
2	16	6	4
3	16	77	2
4	17	40	4

A cluster scatter plot and an elbow-method WCSS plot are generated.